

EDUVISION

HIGHER EDUCATION PERFORMANCE DASHBOARD

Project Documentation

Infosys Springboard

2026

1. Project Overview

EduVision – Higher Education Performance Dashboard is an interactive data visualization project developed to analyse higher-education performance using university ranking and institutional data. The project combines data collection, data cleaning, KPI engineering and Tableau visualization into a unified dashboard solution. It enables users to explore university performance, research activity, student-related indicators and country-level comparisons through interactive dashboards.

2. Problem Statement

Higher-education datasets contain a large number of indicators related to university rankings, research, students, faculty and internationalization. Analysing these indicators separately can make comparison difficult and time-consuming. The EduVision project addresses this challenge by preparing and transforming the data and presenting the important indicators through interactive Tableau dashboards, allowing users to understand and compare higher-education performance more effectively.

3. Objectives

- To collect and prepare university ranking and institutional datasets.
- To clean, standardize and transform the collected data.
- To generate meaningful performance KPIs for analysis.
- To develop interactive dashboards using Tableau.
- To analyse university, research, student and country-level performance.
- To integrate the dashboards into a unified visualization solution.

4. Dataset Description

The project uses university ranking and institutional datasets containing information related to university rankings, academic performance, research, citations, students, faculty, internationalization and other institutional indicators. The primary data sources include **QS 2024 Rankings** and **Times Higher Education (THE) / world ranking data**. The datasets were collected, cleaned and transformed before being used for KPI generation and Tableau dashboard development.

Main categories of data include:

- University and ranking information
- Overall and academic performance scores
- Research and citation indicators
- Student and faculty information
- International student and faculty indicators
- Country and regional information

5. Tools and Technologies

Technology / Tool	Purpose
Python	Data collection, cleaning and transformation
Pandas	Data manipulation and preprocessing
CSV / Excel	Data storage and intermediate processing
Tableau	Interactive dashboard development and visualization
Git	Version control
GitHub	Project repository and collaboration

6. Project Methodology

The project follows a structured data-to-dashboard workflow. University ranking data was first collected from the selected sources. The collected datasets were then cleaned and standardized to improve consistency and usability. Required KPIs were subsequently generated from the prepared data. A dashboard storyboard was created to plan the visualizations, after which the dashboards were developed in Tableau. Finally, the four dashboards were integrated into a unified dashboard solution.

Data Collection → Data Cleaning → KPI Engineering → Storyboarding → Dashboard Development → Dashboard Integration

7. Project Implementation

The project was implemented through six modules covering the complete workflow from data collection to dashboard integration. Each module contributed to preparing the final interactive higher-education performance dashboard.

Module	Implementation
Module 1 – Data Collection	Collected and prepared the QS 2024 and world/THE ranking datasets required for the project.
Module 2 – Data Cleaning and Transformation	Cleaned, standardized and transformed the collected data to improve consistency and usability for analysis.
Module 3 – KPI Engineering	Generated the required performance indicators and prepared the final analytical dataset for dashboard development.
Module 4 – Dashboard Planning	Created the dashboard storyboard to plan the layout, KPIs, charts, filters and overall user flow.

Module 5 – Dashboard Development	Developed the University Overview and Research Analytics dashboards in Tableau.
Module 6 – Dashboard Integration	Developed the Student Analytics and Country Comparison dashboards and integrated all four dashboards into the final solution.

8. Project Structure

The project files were organized according to the implementation modules to maintain a clear separation between data preparation, analysis, planning and dashboard deliverables.

Module_1

- Data collection script
- QS ranking dataset
- Raw university data
- World ranking dataset

Module_2

- Data cleaning script
- Cleaned university dataset

Module_3

- KPI generation script
- Final analytical dataset

Module_4

- Dashboard storyboard

Module_5

- University Overview and Research Analytics Tableau workbook

Module_6

- Final integrated Tableau workbook containing all four dashboards

9. Dashboard Development

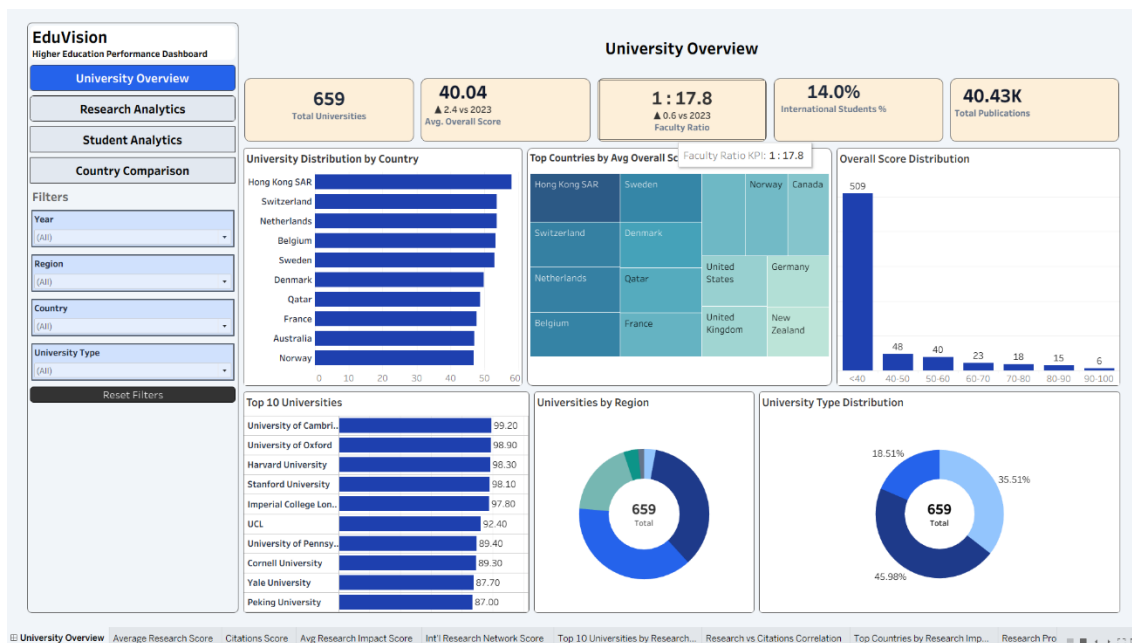
The final Tableau solution consists of four interconnected dashboards, each focusing on a different aspect of higher-education performance. The dashboards use KPI cards, charts, maps, filters and comparison visualizations to provide an interactive analytical experience.

9.1 University Overview

The University Overview dashboard provides a consolidated view of university performance. It presents key indicators such as the total number of universities, average overall score, faculty-to-student ratio, international student percentage and total publications. The dashboard also enables analysis of university distribution by country, overall score distribution, top-performing universities, regional composition and university type.

Key KPIs visible in the dashboard:

- Total Universities: **659**
- Average Overall Score: **40.04**
- Faculty Ratio: **1:17.8**
- International Students: **14.0%**
- Total Publications: **40.43K**



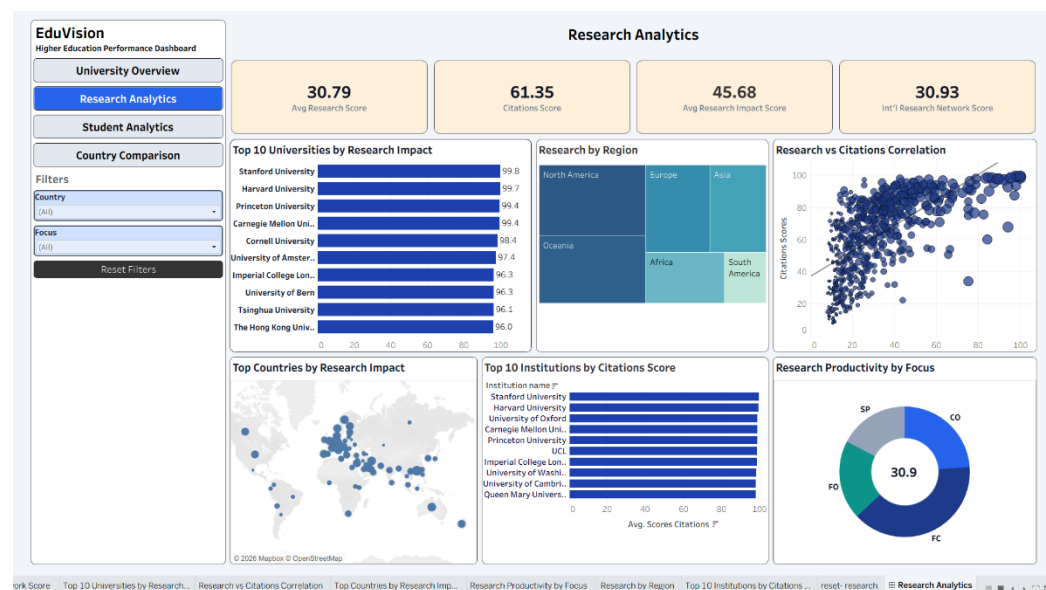
9.2 Research Analytics

The Research Analytics dashboard focuses on research-related performance across universities. It presents research score, citation score, research impact and international research network indicators. The dashboard also provides comparisons of top universities and institutions, regional research distribution, country-level research impact, research-citation relationships and research productivity by focus.

Key KPIs visible in the dashboard:

- Average Research Score: **30.79**
- Citations Score: **61.35**
- Average Research Impact Score: **45.68**

- International Research Network Score: **30.93**

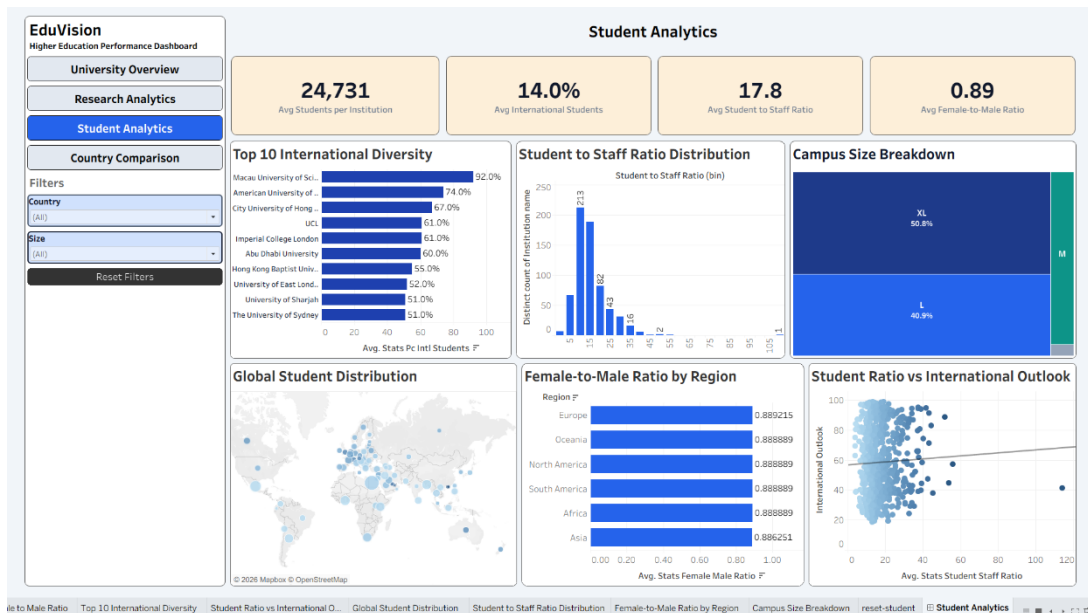


9.3 Student Analytics

The Student Analytics dashboard focuses on student-related characteristics across universities. It presents indicators for student population, international students, student-to-staff ratio and female-to-male ratio. The dashboard also includes visualizations for international diversity, student-to-staff ratio distribution, campus size, global student distribution, regional gender ratios and the relationship between student ratio and international outlook.

Key KPIs visible in the dashboard:

- Average Students per Institution: **24,731**
- Average International Students: **14.0%**
- Average Student-to-Staff Ratio: **17.8**
- Average Female-to-Male Ratio: **0.89**



9.4 Country Comparison

The Country Comparison dashboard provides a country-level view of higher-education performance. It enables comparison of countries using international outlook, student-to-staff ratio, regional distribution and multiple performance indicators. The dashboard combines ranking, geographical and comparative visualizations to support country-level benchmarking.

Key KPIs visible in the dashboard:

- Top Ranked Country: **Hong Kong SAR**
- Global Average Score: **44.5**
- Average International Faculty Score: **83.6**
- Average Student-to-Staff Ratio: **19.1**



10. Key Features

The EduVision dashboard provides the following key features:

- Interactive KPI cards for quick performance analysis.
- Filters for year, region, country, university type and other relevant dimensions.
- University ranking and performance comparisons.
- Research and citation performance analysis.
- Student and international diversity analysis.
- Country and regional benchmarking.
- Geographical visualizations for global comparison.
- Interactive reset controls for returning to the default view.
- Integrated navigation across all four dashboards.

11. Testing and Validation

The dashboards were tested to ensure that the analytical results and interactive features function correctly. KPI values were checked against the prepared dataset, while filters, reset controls, dashboard navigation and visualizations were reviewed for proper interaction. The final Tableau workbook was also checked to ensure that all four dashboards were integrated successfully.

12. Key Outcomes

The completed EduVision solution converts university ranking and institutional data into an interactive analytical dashboard. It provides a unified view of overall university performance, research activity, student characteristics and country-level performance. The dashboards make it easier to compare institutions and identify differences across countries and regions.

13. Challenges Faced

During the development of the project, the major challenges involved handling data from multiple sources, maintaining consistency between university records, preparing suitable KPIs and designing interactive dashboards with clear and readable visualizations. These challenges were addressed through systematic data cleaning, transformation and iterative dashboard development.

14. Future Scope

The EduVision dashboard can be further enhanced by incorporating additional ranking years and institutional indicators. Historical trend analysis, advanced comparative analytics and predictive features can also be introduced to provide deeper insights into higher-education performance.

15. Conclusion

EduVision successfully combines data collection, data preparation, KPI engineering and interactive Tableau visualization into a unified higher-education performance dashboard. The final solution provides four interconnected analytical views covering university performance, research analytics, student analytics and country comparison. The project demonstrates how data visualization can transform complex institutional data into a clear and interactive analytical solution.